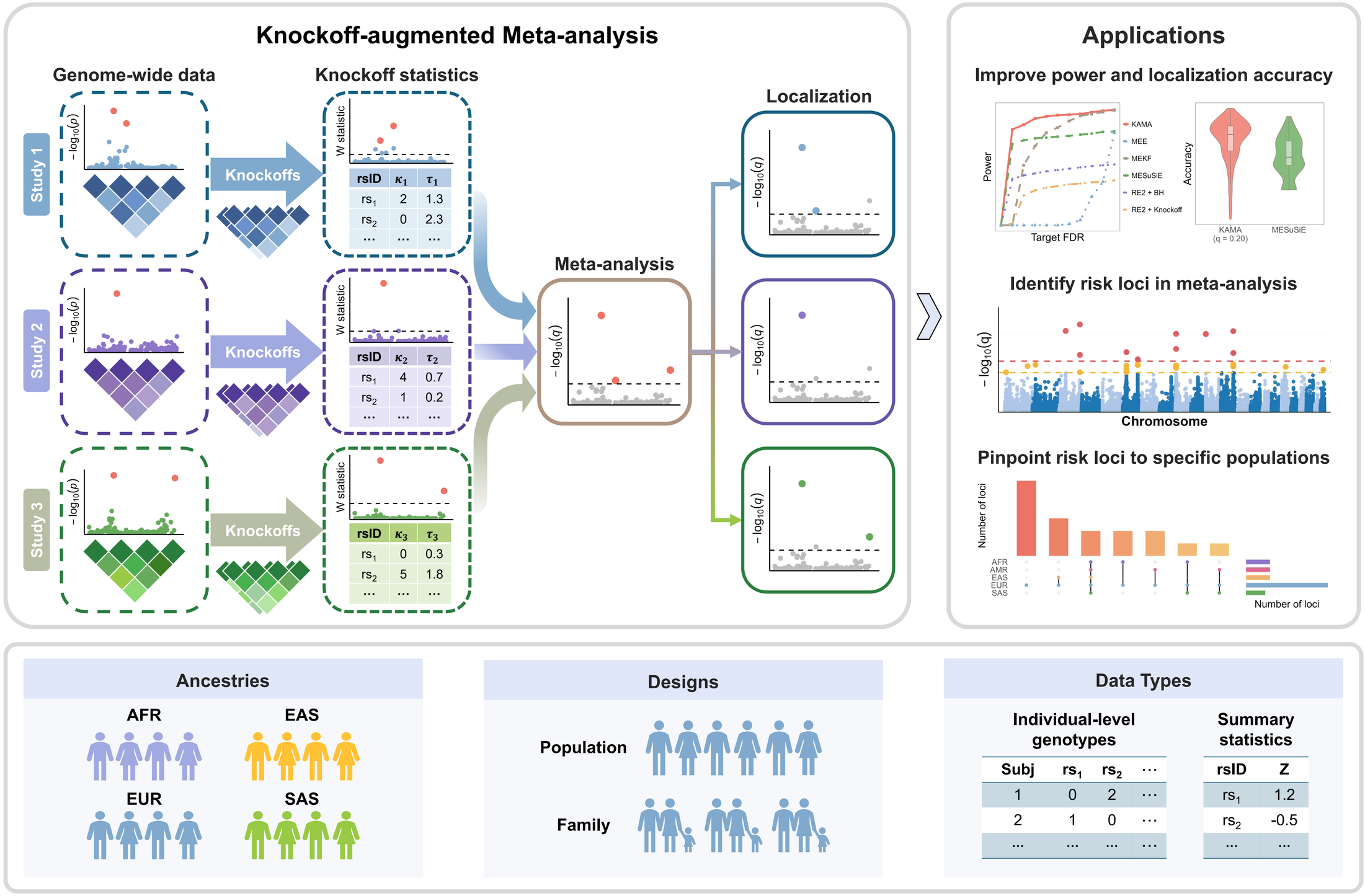
Knockoff-augmented Meta-analysis
Genome-wide data
Knockoff statistics
Study 1
Study 2
Study 3
Knockoffs
W statistic
rsID κ_1 τ_1
rs$_1$ 2 1.3
rs$_2$ 0 2.3
rsID κ_2 τ_2
rs$_1$ 4 0.7
rs$_2$ 1 0.2
rsID κ_3 τ_3
rs$_1$ 0 0.3
rs$_2$ 5 1.8
Meta-analysis
Localization
$-\log_{10}(p)$
$-\log_{10}(q)$
Applications
Improve power and localization accuracy
Power
Target FDR
KAMA
MEE
MEKF
MESuSiE
RE2 + BH
RE2 + Knockoff
Accuracy
KAMA (q = 0.20)
MESuSiE
Identify risk loci in meta-analysis
Chromosome
Pinpoint risk loci to specific populations
Number of loci
AFR
AMR
EAS
EUR
SAS
Ancestries
AFR
EAS
EUR
SAS
Designs
Population
Family
Data Types
Individual-level genotypes
Subj rs$_1$ rs$_2$ ⋯
1 0 2 ⋯
2 1 0 ⋯
Summary statistics
rsID Z
rs$_1$ 1.2
rs$_2$ -0.5